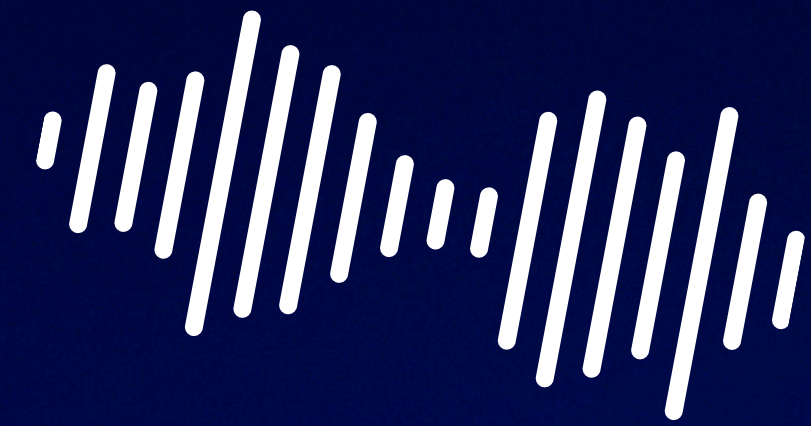
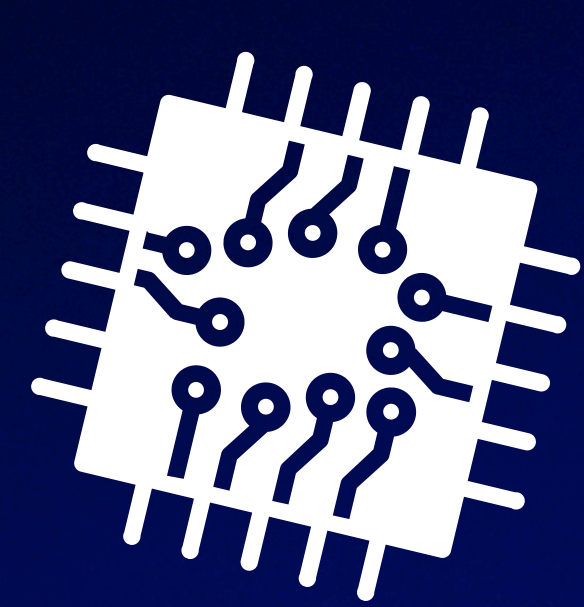
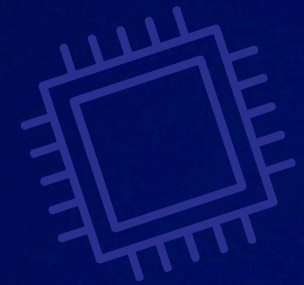


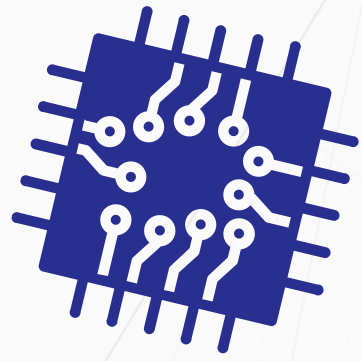


DESIGN & STIMULATION OF A SMART WATER TANK OVERFLOW PROTECTION & PUMP CONTROL SYSTEM

Presented by Group 10

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KARTHIKEYA (2620040024)
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SNIGDHA (2620040042)
AISHWARYA (2620040059)





CORE SYSTEM OBJECTIVES

OVERFLOW PREVENTION

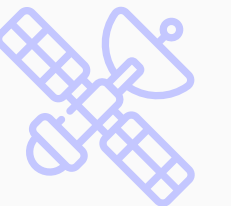
Eliminate water spillage, structural dampness, and energy waste by providing automated high-limit power disconnection.

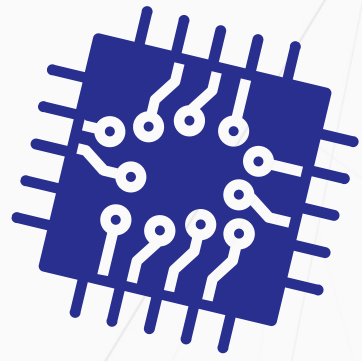
HARDWARE PROTECTION

Prevent motor burnout caused by dry running or incessant power rapid-cycling near critical operational thresholds.

DETERMINISTIC RELIABILITY

Achieve sub-millisecond response using ultra-low-cost, zero-software combinational NOR logic gates rather than complex MCUs



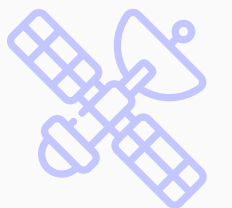


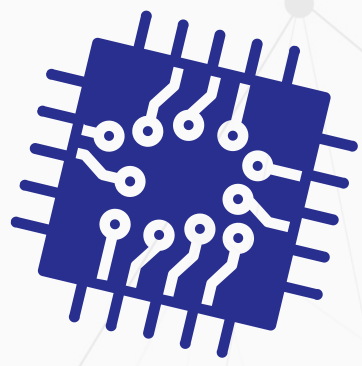
SYSTEM INPUTS & SENSING

Low Sensor (SL): Placed near tank floor. Outputs logic 1 when water touches low boundary.

- Medium Sensor (SM): Positioned at standard target capacity. Triggers primary pump shutoff when submerged

. - High Sensor (SH): Positioned at capacity ceiling. Serves as redundant fail-safe overflow protection. - Interface Circuitry: Conductive level probes connected to $10k\Omega$ pull-down resistors for clean digital logic states.





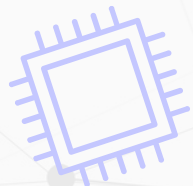
CONTROL OUTPUT & ACTUATION

SINGLE OUTPUT CONTROL (P)

THE CONTROL DECISION DRIVES A SINGLE BINARY OUTPUT SIGNAL P TO CONTROL THE AC PUMP MOTOR SWITCH.

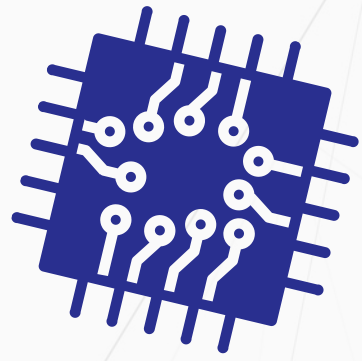
$P = 1$ (PUMP ON): ACTIVATED WHEN WATER LEVEL DROPS BELOW REQUIRED OPERATIONAL THRESHOLDS.

$P = 0$ (PUMP OFF): IMMEDIATELY TURNS OFF THE RELAY ONCE TARGET CAPACITY IS DETECTED. $P=SH+SM$



Input-Output Logic Mapping

HIGH (s_H)	MID (s_M)	LOW (s_L)	PUMP (P)	SYSTEM OPERATIONAL CONDITION
0	0	0	1 (ON)	Tank completely empty below low level → Start Refill
0	0	1	1 (ON)	Water between Low and Mid → Continue Refilling
0	1	1	0 (OFF)	Target operational level reached → Stop Pump
1	1	1	0 (OFF)	High capacity overflow limit → Lock Pump OFF
Other	Combinations	-	X (OFF)	Physically impossible sensor states treated as Don't Care (X)



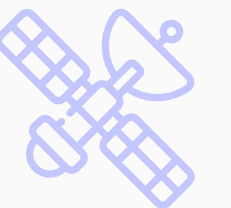
Control Output

We have one output called P.

When P is 1, the pump is ON, and the tank can fill.

When P is 0, the pump is OFF, so the water stops filling when the required level is reached.

In simple terms, the system checks the water level, decides whether to run, and automatically switches the pump ON or OFF.



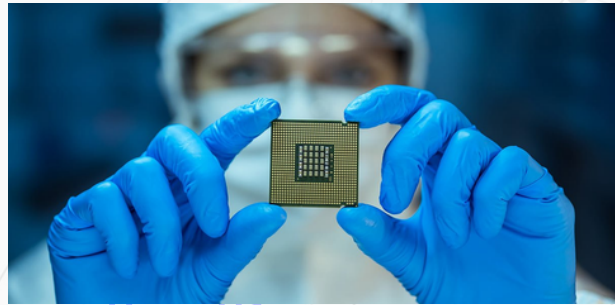
$P = \frac{SH \cdot SM}{SL + SH \cdot SM} \cdot SL$



IMPLEMENTATION OF LOGIC GATES

Required Integrated Circuits (ICs) To build this exact circuit, you need three low-cost digital logic ICs:

- 74LS04 (Hex Inverter): Contains 6 independent NOT gates (used to invert sensor inputs).
- 74LS32 (Quad 2-Input OR): Contains 4 independent OR gates (used for the feedback latch).
- 74LS08 (Quad 2-Input AND): Contains 4 independent AND gates (used to combine the logic).



Step 1: Inverting the High Sensor:- Connect the High Sensor output line to 74LS04 Pin 1 (Input A1). 74LS04 Pin 2 (Output Y1) now delivers the inverted signal.

Step 2: Creating the Feedback Loop:- (OR Gate) Connect the Low Sensor output line to 74LS32 Pin 1 (Input A1). Connect 74LS32 Pin 2 (Input B1) to the final output of the circuit (74LS08 Pin 3). This is the crucial feedback wire that creates memory. 74LS32 Pin 3 (Output Y1) now delivers the combined term.

Step 3: Combining with the Interlock:- (AND Gate) Connect 74LS32 Pin 3 (OR Output) to 74LS08 Pin 1 (Input A1). Connect 74LS04 Pin 2 (NOT Output) to 74LS08 Pin 2 (Input B1). 74LS08 Pin 3 (AND Output) is your final Pump Control Signal.



KEY SMART FEATURES

Zero Overflow

Automatically detects when the tank reaches its maximum level.
The pump switches OFF to prevent overflow and water wastage.

DRY-RUN PROTECTION

DETECTS INSUFFICIENT WATER AT THE SOURCE BEFORE RUNNING THE PUMP.
PREVENTS DRY RUNNING AND PROTECTS THE PUMP FROM DAMAGE.

INSTANT VISUAL ALERT

LEDs OR A DISPLAY PROVIDE IMMEDIATE INFORMATION ABOUT THE TANK STATUS.
USERS CAN QUICKLY IDENTIFY CONDITIONS LIKE LOW, FILLING, FULL.



ZERO OVERFLOW

Automatically detects when the tank reaches its maximum level.
The pump switches OFF to prevent overflow and water wastage.



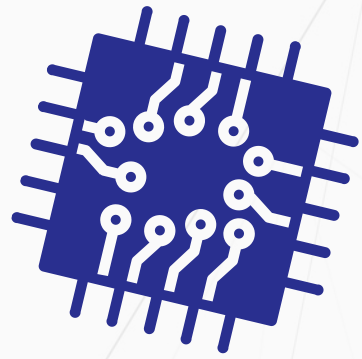
DRY-RUN PROTECTION

Detects insufficient water at the source before running the pump.
Prevents dry running and protects the pump from damage.



INSTANT VISUAL ALERT

LEDs or a display provide immediate information about the tank status.
Users can quickly identify conditions like Low, Filling, Full, or Error.

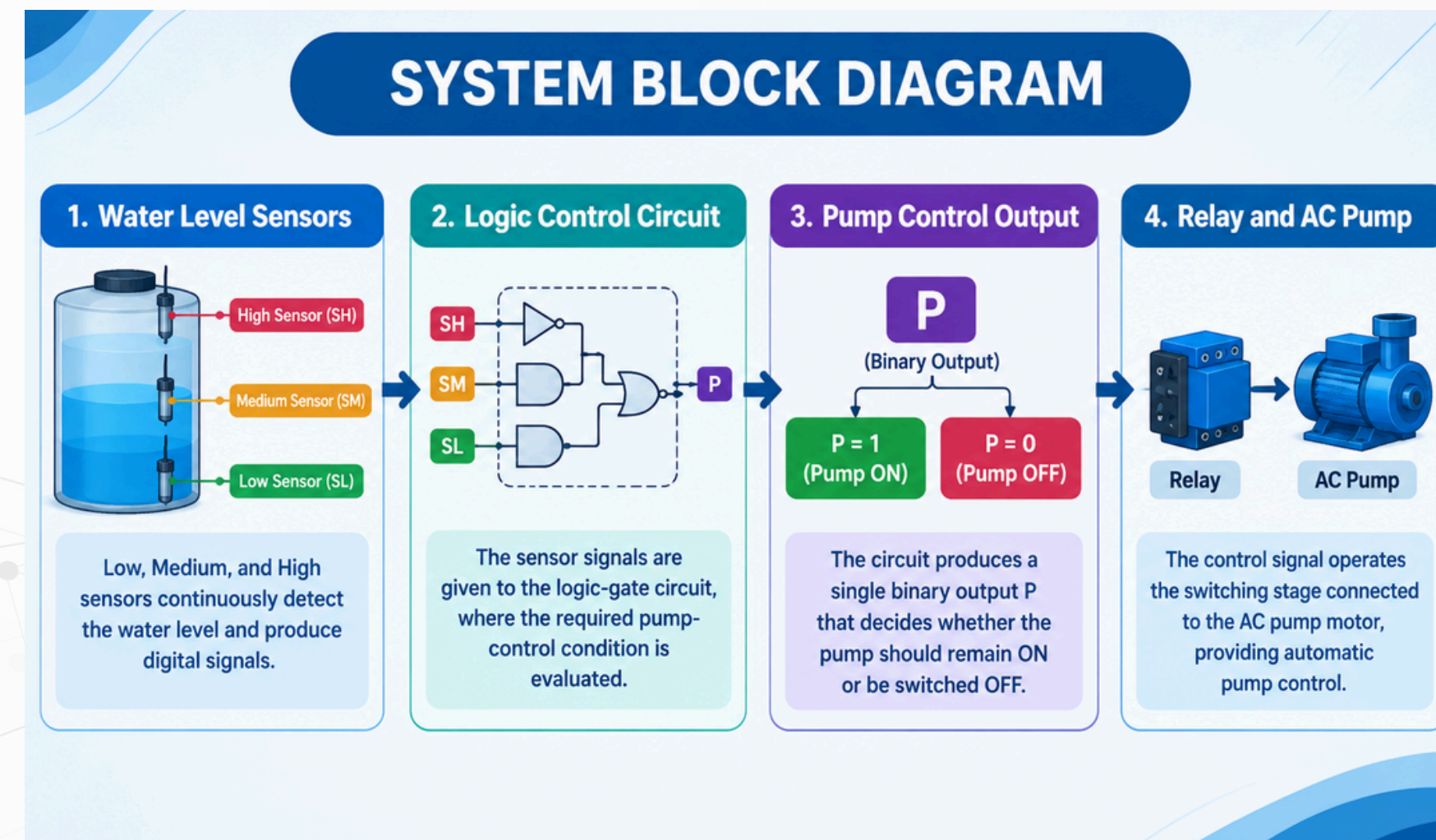


SYSTEM BLOCK DIAGRAM

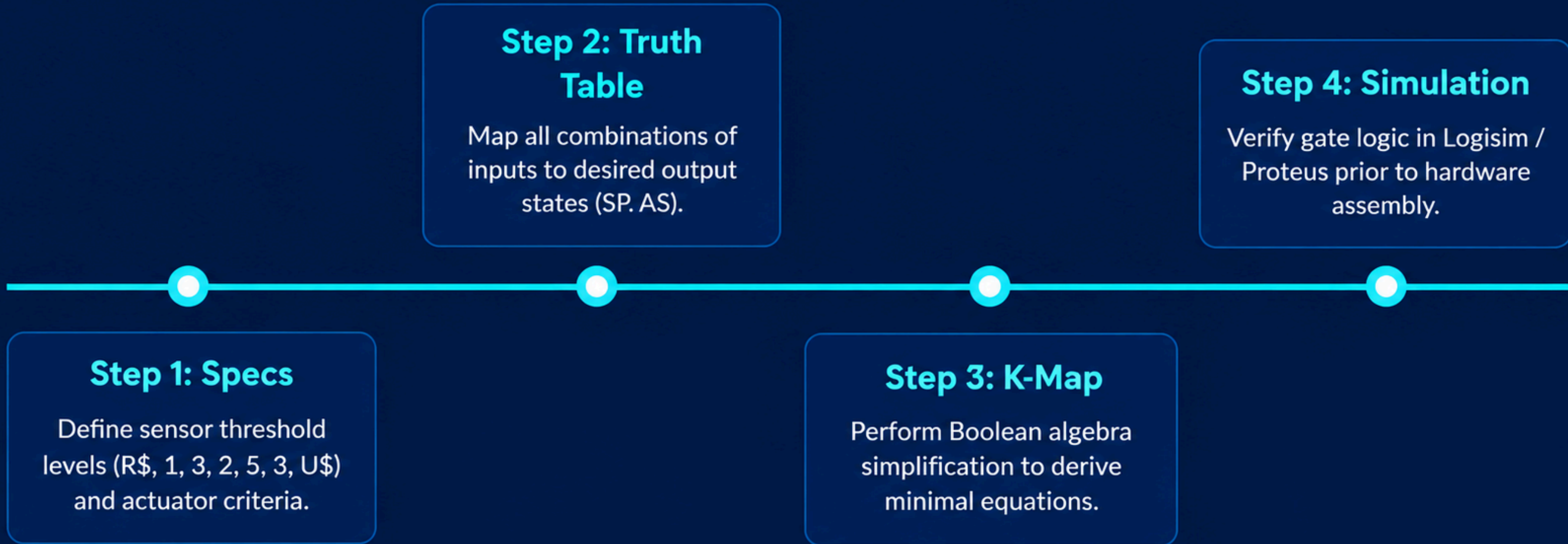
Water Level Sensors: Low, Medium, and High sensors continuously detect the water level and produce digital signals.

Logic Control Circuit: The sensor signals are given to the logic-gate circuit, where the required pump-control condition is evaluated.

Relay and AC Pump: The control signal operates the switching stage connected to the AC pump motor, providing automatic pump control.



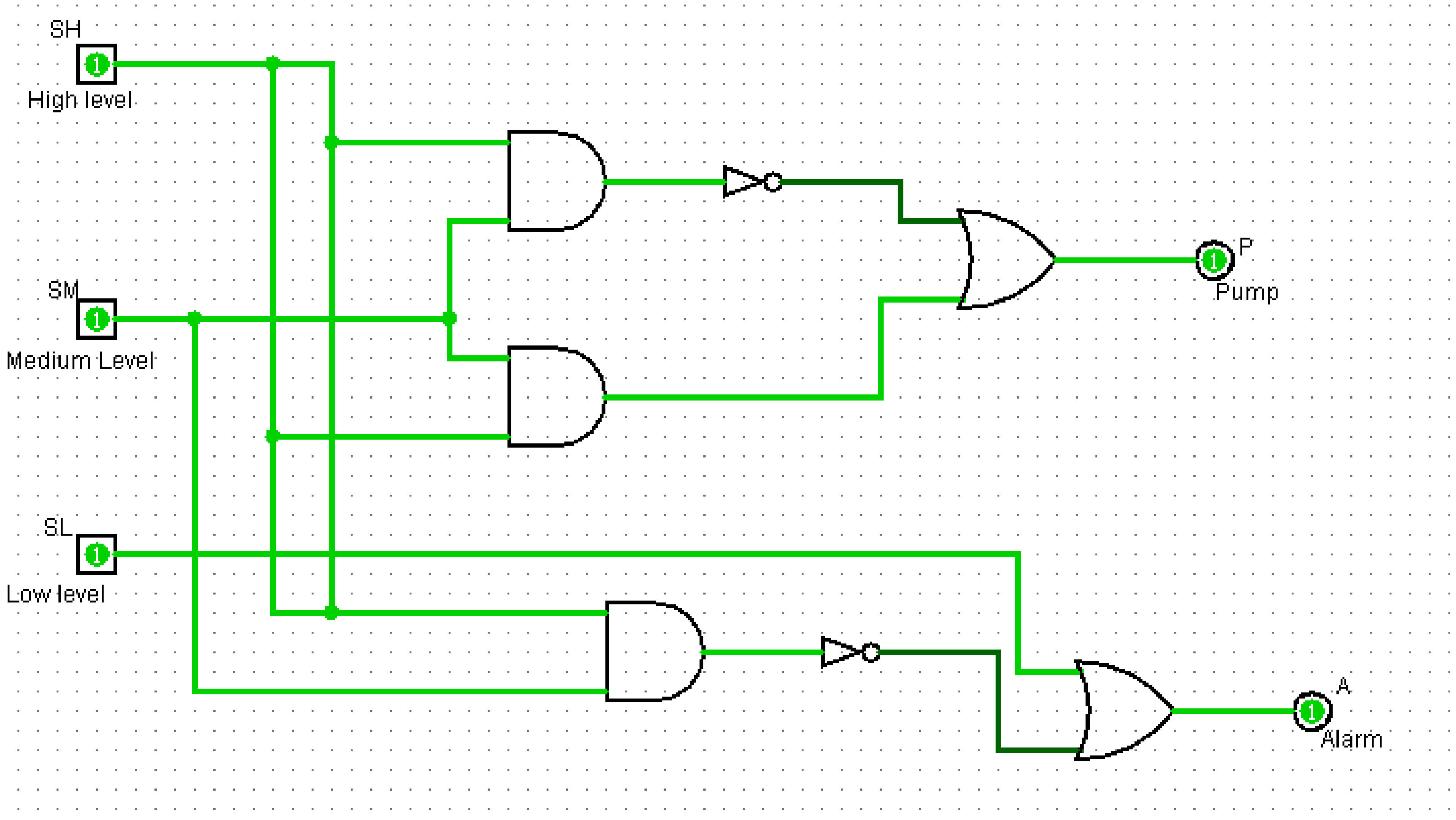
Design & Simulation Workflow

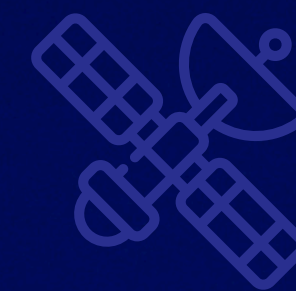
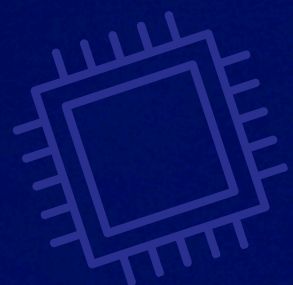
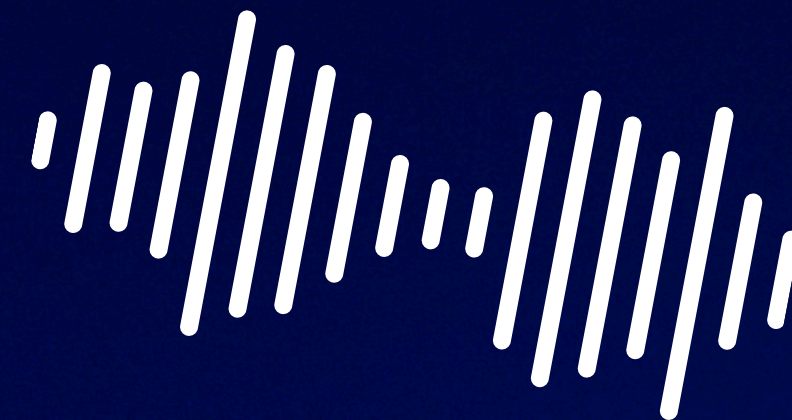
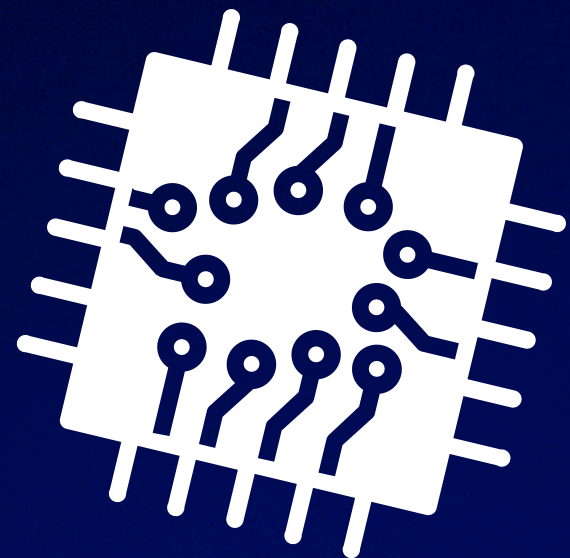


System Efficiency Comparison



Combinational logic control eliminates human latency and overflow events, significantly enhancing motor life and resource conservation compared to manual control.





THANK
YOU

